Positive Correlation:

The correlation is said to be positive if the values of two variables move in the same direction.

Ex 1: If income and Expenditure of a Household may be increasing or decreasing simultaneously. If so, there is positive correlation. Ex. Y= a + bx

Negative Correlation:

The Correlation is said to be negative when the values of variables move in the opposite directions. Ex. Y= a – bx

Ex 1: Price and demand for a commodity move in the opposite direction.

Type II: Based upon the number of variables studied.

There are three types based upon the number of variables studied as

a) Simple Correlation
b) Multiple Correlation
c) Partial Correlation

Simple Correlation:

If only two variables are taken for study then it is said to be simple correlation.

Multiple Correlations:

If three or more than three variables are studied simultaneously, then it is termed as multiple correlation.

Ex: Determinants of Quantity demanded

$$Q_d = f(P, Pc, Ps, t, y)$$

Where Q_d stands for Quantity demanded, f stands for function.

P is the price of the goods,

P_c is the price of competitive goods

P_s is the price of substituting goods

t is the taste and preference

y is the income.

Partial Correlation:

If there are more than two variables but only two variables are considered keeping the other variables constant, then the correlation is said to be Partial Correlation

Type III: Based upon the constancy of the ratio of change between the variables

Correlation is divided into two types as linear correlation and Non-Linear correlation based upon the Constancy of the ratio of change between the variables.

Linear Correlation: Correlation is said to be linear when the amount of change in one variable tends to bear a constant ratio to the amount of change in the other.

Ex. Y= a + bx

Non Linear: The correlation would be non-linear if the amount of change in one variable does not bear a constant ratio to the amount of change in the other variables.

Ex. $Y = a + bx^2$

Methods of Studying Correlation:

The various methods of ascertaining whether two variables are correlated or not are:

1. Scatter diagram Method
2. Graphic Method
3. Karl Pearson's Co - efficient of correlation and
4. Method of Least Squares.

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